

Kellen Cheng

✉ kellentan@alumni.princeton.edu ☎ 4074094268 🏠 Website 📧 kellentan 🇺🇸 U.S. Citizen

Education

Princeton University

Aug. 2022 - Nov. 2024

M.A. ELECTRICAL AND COMPUTER ENGINEERING, GPA: 4.000

PHD CANDIDATE (ON-LEAVE), SPECIALIZATION: NATURAL LANGUAGE PROCESSING

PRINCETON, NJ

University of California, Los Angeles (UCLA)

Sept. 2018 - Jun. 2022

B.S. ELECTRICAL ENGINEERING, TECHNICAL BREADTH COMPUTER SCIENCE, GPA: 3.926

LOS ANGELES, CA

Experience

Microsoft: Applied Scientist II

Oct. 2025 - Present

MANAGER: DR. MINGXI CHENG

MOUNTAIN VIEW, CA

- Data curation, generation, and RL post-training for multimodal foundation models as part of the PPT ML team.

IBM Research: AI Research Scientist Intern

May. 2025 - Aug. 2025

MENTOR: DR. ANNA LISA GENTILE

SAN JOSE, CA

- Formulated a method to dynamically generate type-constrained negative samples to enhance an LLM's factuality.
- Constructed an automatic pipeline which exploits structured source data, generating synthetic data which is used to seed longer context generation.
- Designed a framework to better embed misinformation into negative samples by utilizing GRPO along with an RL self-play custom-defined reward function.
- First-author work in submission; two patents (one on fine-tuning via multiagent debate traces, another on type-constrained adversarial factuality for proprietary LLMs) filed for disclosure.

Samsung Research America: NLP Research Scientist Intern

Jan. 2025 - May. 2025

MENTOR: DR. GANESH RAMESH

MOUNTAIN VIEW, CA

- Designed a multi-agentic LLM framework to improve conversation summarization through iterative text feedback.
- Adapted the framework for on-device local inference with Apple Silicon using Ollama and MLX-LM.
- Ran quantized LoRA (QLoRA) instruction fine-tuning for a range of language models spanning 0.5B to 9B parameters.
- First-author work accepted as an EACL 2026 oral paper.

IBM Research: AI Research Scientist Intern

Jun. 2024 - Sept. 2024

MENTOR: DR. ANNA LISA GENTILE

SAN JOSE, CA

- Synthesized and curated an evaluation benchmark for health advice guardrails from Common Crawl web text.
- Designed and implemented a sparse human-in-the-loop system for semi-automatic annotation of synthetic data at scale.
- Formulated a method to automatically generate synthetic data using compact LLMs for health advice guardrails.
- Fine-tuned scalable and compact detector models on a blend of synthetic and open-source training data, beating GPT-4o by 3.73% in accuracy and 1.54% in F1-score, despite containing 400x less parameters.
- Created an internal Rest API that integrated my detector model and automated internal model evaluations for the team.
- First-author works published in EMNLP 2024 industry track, AACL 2025 main conference; one patent (continual learning and knowledge distillation framework for AI safety guardrail development) filed for disclosure.

Princeton: NLP Researcher

Nov. 2022 - Nov. 2024

ADVISOR: DR. SUMA BHAT

PRINCETON, NJ

- Created an end-to-end two-mask infilling fine-tuning objective for idiomatic knowledge injection using the IEKG dataset.
- Implemented two-stage fine-tuning with transfer learning to achieve new state-of-the-art performance of 83.75% accuracy on the IMPLI benchmark, an improvement of 12% compared to previous state-of-the-art.

- Conducted ablation and data perturbation studies to gauge contextual reasoning capabilities for off-the-shelf language models ranging from 0.5B to 7B parameters, uncovering that they actually perform *better* without the context.
- Works published in EMNLP 2023 main conference (second-author) and NAACL 2024 main conference (first-author).

Selected Publications

- **Kellen Tan Cheng**, Ganesh Ramesh, Nafiu Rashid, Geoffrey Jay Tso, Jilong Kuang. *Compact Language Models with Iterative Text Refinement for Health Dialogue Summarization*. EACL 2026 Main (Oral).
- **Kellen Tan Cheng**, Anna Lisa Gentile, Chad DeLuca, Guang-Jie Ren. *STAR: Self-Automated Back-Querying for Production Data Generation*. IJCNLP-AAACL 2025 Main.
- Chad DeLuca, Anna Lisa Gentile, Shubhi Asthana, Bing Zhang, Pawan Chowdhary, **Kellen Tan Cheng**, Basel Shbita, Pengyuan Li, Guang-Jie Ren, Sandeep Gopisetty. *OneShield - the Next Generation of LLM Guardrails*. ArXiv 2025.
- **Kellen Tan Cheng**, Anna Lisa Gentile, Pengyuan Li, Chad DeLuca, Guang-Jie Ren. *Don't Be My Doctor! Recognizing Healthcare Advice in Large Language Models*. EMNLP 2024 Industry Track.
- **Kellen Tan Cheng**, Suma Bhat. *No Context Needed: Contextual Quandary In Idiomatic Reasoning With Pre-Trained Language Models*. NAACL 2024 Main.
- Ziheng Zeng, **Kellen Tan Cheng**, Srihari Venkat Nanniyur, Jianing Zhou, Suma Bhat. *IEKG: A Commonsense Knowledge Graph for Idiomatic Expressions*. EMNLP 2023 Main.

Awards & Organizations

Toby & Jack Wolf Travel Grant	2024
Bede Liu Travel Grant	2023
Princeton ECE Departmental Fellowship	2022
Tau Beta Pi	2020 - Present
IEEE Eta Kappa Nu (HKN)	2019 - Present
UCLA Dean's Honor List	2019 - 2022

Skills

Languages	Python, C++, MATLAB
Frameworks	PyTorch, Tensorflow, MLX, Transformers, TRL
Tools	Ollama, Slurm, LSF, Azure, AWS EC2, Conda, Jupyter, LaTeX, Git, WSL, MS Office